Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/12/2012
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Microgravity and Magnetic Fields - Directional Root Growth and Gene Expression		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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This proposal combines morphological measurements on micro-gravity germinated and grown (Brassica) roots that are exposed to high-gradient magnetic fields (HGMF) with a high-resolution spatial analysis of genes involved in gravitropism. Magnetic gradients exert a repulsive force on diamagnetic substances such as the starch in amyloplasts, which results in gravitropic-like curvature. The proposal is based on hardware that was developed for a previous flight experiment on STS-107. Like in the originally proposed experiment, microscopic analysis of amyloplast size and distribution will be studied in the context of achieved curvature. In addition, high-resolution genetic profiling will be conducted that focuses on the cytoskeleton and starch development. The result of the proposed work will enable directional growth of seedlings and eventually mature plants under microgravity conditions using HGMF. Three conditions will be used to differentiate microgravity from uniform magnetic fields and HGMF conditions. Only HGMF

Task Description:	is expected to influence root curvature. A uniform magnetic field will serve as control for possible secondary magnetic effects. This condition will be compared to an in-flight control without magnetic field to determine growth rate, gene expression, growth, and directionality (autotropism). Root material from these experiments will be used for investigations on cytoskeletal organization and the expression of starch-related genes. A second aspect of the proposed work will investigate the process of gravisensing, whether dependent on amyloplast position, movement, or force exerted onto amyloplasts. Space experimentation is necessary to establish a threshold value for a directional magnetic force. In addition to providing insight in the fundamental mechanism of the gravisensing system of plants, the data from this experiment can be used to develop technology to provide a directional, gravity-like stimulus to plants during early stages of germination. The proposed experiment can be performed with existing hardware, be adapted easily to changing flight schedules, and requires minimal crew time. The anticipated benefits include a better understanding of root development, effects of amyloplast displacement on root development and curvature, and compensatory measures to alleviate microgravity conditions. A newly developed gene extraction technology is ideally suited for gene-profiling studies and has far-reaching implications for studying gene expression, especially for confined space flight experiments. Additional benefits of the project include training of students at the undergraduate and graduate levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to an improved understanding of basic biological questions, such as identification of the site of graviperception, curvature-induced (but not gravity-dependent) gene expression, the experiment also tests the effect of magnetic fields on plant roots. Therefore, biological questions are investigated in conjunction with physical principles. We intent to analyze the effect of static magnetic fields on biological systems. This work will provide a starting point for the current discussion of the detrimental effects of (electro)magnetic fields. In addition, understanding basic biological questions will improve the prospect for enhanced plant productivity, resource optimization, and reduced maintenance.
Task Progress:	To accomplish the first-year objectives, we formed a research, advisory and development team. The team consists of the support group at KSC (Howard Levine, Ralph Fritsche, Charlie Quincy, Trevor Murdoch, Natasha Rapp, David Reed), and the PI team (Karl H. Hasenstein, Andrea Edge (graduate student), and Myoung Roul Park (post-doctoral scientist). Dr. Park joined the PI team March 2011. The team addressed three main objectives: 1) Biological Studies ; 2) Hardware development ; 3) Development of experimental protocols for ground studies and flight experiment. Biological studies: We determined germination requirements, seed adhesion, imbibition duration, germination, temperature sensitivity of root elongation, and fixation conditions. Growth tests were started immediately after the grant was initiated. We used existing Magnetic Field Chambers (MFCs) and Seed Cassettes (legacy design) at the PI lab. The tests were continued once modified seed cassettes became available. The required water volume for germination was examined and we found that a single dispensation of 400 µl water produced optimal germination and growth provisions. This volume was sufficient to provide moisture for the entire growth duration and supported growth for a root length of more than 5 cm, a size greater than what was required for the projected experiment. Demonstration of seed germination after imbibing two seed/paper sandwiches with 400 µl water. The resulting images illustrate uniform germination, root hair formation, and root extension between paper as a result of gravitropic curvature of those roots that germinated in an inverted orientation. Seed Adhesion: The small seed size and their uniform round shape make precise positioning of the seeds very difficult without proper and reliable adhesion of seeds to the substrate. We tested guar gum, gelatin, guar/gelatin mixtures, and polyvinyl alcohol (PVA) as well as guar gum and PVA mixtures as adhesives. The most reliable adhesion that did not negatively affect seed germination was obtained with 20% PVA (Mowiol) and commercially available Elmer's School Glue. This compound dries clear, is non-toxic, and eventually water soluble ('washable'). Seeds attached with this glue showed no measurable delay in germination, no reduced growth rate, but the seed coat remained attached to the germination paper better than any other tested glue. In addition, the placement of the seeds toward the center of the paper sandwich, about 1 mm away from the paper edge provided greater reliability because emerging root hairs interact with the paper and further stabilize seed attachment. Fixation in seed cassettes inside MFCs with 70% ethanol or aqueous buffers with formaldehyde and extensive manual shaking did not dislodge seeds. Despite the delay in obtaining new (MFCs) with properly spaced wedges, Teflon coating, and sealed covers delayed fixation studies but temporary fixes and necessary biocompatibility tests showed that seed germination is robust in a regular lab system (Petri dishes) and MFCs. Storing seeds in MFCs for 3 months did not reduce subsequent germination rate or cause slower root growth. Temperature dependency of germination and root elongation was tested and resulted in a maximum growth rate at 30C. Lower and higher temperatures reduced root growth. A combined algorithm was developed that describes root growth as a sum of temperature and time with the following equation: Growth = (-0.0033 {temp}^2+0.191{temp}-2.15)×{time} + 0.36×{temp} -19 This relationship explains ~90% of the experimental variability; the coefficients will be further refined as more experiments will be conducted. Cytoskeletal Staining Protocols: One of the objectives of the flight experiment is the determination of the influence of microgravity on cell growth, and the cytoskeletal organization. Therefore a reliable staining protocol had to be developed that takes into account the peculiar circumstances of the experiment (preparation of fixative, the duration for which reliable fixation can be achieved) and the ability to store fixed tissue in fixative without negatively affecting subsequent staining procedures. Therefore, we tested different concentrations of formaldehyde fixative (1, 2, and 4%) and different storage combinations (fixative storage before use and storage of tissue in fixative after fixation). So far, optimal fixation conditions were achieved by keeping fixative for three weeks or less before fixation (at room temperature). Suitable staining could be achieved with root tissue stored up to four weeks in fixative that was stored three weeks prior to fixation. While the schedule is complex and needs to consider time (as limiting factor before formaldehyde polymerizes and therefore loses its ability to cross-link) and temperature, which affects polymerization, a decline in the fixation is not precipitous and does not affect all tissue equally. Microtubule staining was accomplished with primary antibody YOL-1/34 and secondary antibody rabbit anti-rat IgG

	(labeled with Alexa 488). The protocol followed standard lab procedures and resulted in reliable staining. Initial attempts to label F-actin with Rhodamin-phalloidin failed despite multiple trials. We then tested mouse mAB anti-actin Clone C4 with goat anti-mouse IgG labeled with Alexa 546 as secondary antibody. This protocol resulted in good staining of F-actin. Starch staining and identification: The ability to localize starch in columella cells and the possible modification that starch experiences in microgravity conditions prompted an ongoing investigation into the labeling of different starch types that also characterizes starch composition. Incubation of starch in dyes such as Safranin B, Alcian Blue, Basic Fuchsin, or combinations thereof, stain starch and produce fluorescence (lambda>515 nm). Staining of raw starch and Spurr-embedded samples indicate that starch analysis is possible. Gene expression studies: A significant aspect of the proposed research concerns the effect of microgravity and HGMF on gene expression in growing roots. We selected genes that represent auxin transport, starch and carbohydrate metabolism, the cytoskeleton, and reference genes. The quality of the mRNA depends greatly on the storage condition after the flight experiment has been completed. Therefore, we tested the effect of storing tissue under various conditions and for extended times in RNA-Later. We compared commercially prepared RNA-Later with in-house prepared solutions and determined that our own preparation is superior to the commercial fixative In addition to the extraction procedure, we tested the gene expression patterns in extracts. These expression patterns compared two reference genes (ubiquitin and actin-7) to other genes based on the storage condition of the tissue. While the complete representation of all data is beyond the scope of this report, the data clearly indicate that gene sequences degraded at different rates. Hardware development: The difference between the legacy and newly designed cassettes were a 5-channel design to accommodate ten instead of 8 seeds, better connection between cover and base, and larger inner space to accommodate a double germination paper sandwich. The newly designed Magnetic Field Chambers (MFCs) were analyzed for the reproducibility of the magnetic properties. While the spacing of the wedges resulted in an equidistant positioning of the maximum field strength, the intensity was not uniform. However, resulting gradient were similar.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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